AKSHAYA NARAYANA

Data Engineer

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PROFILE SUMMARY

- **5+ years of experience** in designing and implementing scalable data solutions, focusing on extracting, transforming, and loading (ETL) big data.
- Expertise in **big data technologies** like Hadoop, Apache Spark (Scala preferred), Python, Apache Hive, and cloud platforms (AWS, GCP, Azure) to build **batch data pipelines** optimized for performance, fault tolerance, and SLA adherence.
- Extensive hands-on experience in **AWS services** (EC2, Athena, Glue, Lambda, S3, RDS) and **Azure IaaS** (Virtual Networks, Virtual Machines, Auto-Scaling, Traffic Manager) for data processing and automation.
- Skilled in **data modeling** and designing schemas to accommodate evolving data sources, facilitating seamless data joins across datasets.
- Proficient in **orchestrating workflows** using Apache Airflow, including creating idempotent workflows for efficient scheduling and monitoring.
- Strong experience in **SQL querying** for data profiling, optimization, and analysis, with familiarity in **BigQuery** and **Spark SQL**.
- Developed **batch processing solutions** using **Azure Data Factory** and **Databricks** to process large-scale data.
- **Kafka producers/consumers** used to enable seamless **data streaming** integrated with AWS services.
- Hands-on experience with **Snowflake** for integrating and processing nested JSON data, utilizing **SnowSQL** and **SnowPipe**.
- Proficient in creating and optimizing **CI/CD pipelines** in **Jenkins** using pipeline syntax and Groovy.
- Experience with **production support** and monitoring tools like **Splunk** and **AWS CloudWatch** for troubleshooting and ensuring system reliability.

TECHNICAL SKILLS

- **Cloud Platforms:**

Azure: Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure Data Lake Storage, Azure SQL Database
AWS: AWS Glue, Amazon Redshift, Amazon S3, AWS Lambda, Amazon RDS, AWS Step Functions, Amazon DynamoDB
GCP: Google BigQuery, Google Cloud Storage, Google Cloud Dataflow, Google Cloud Pub/Sub, Google Cloud SQL

- **Data Engineering & ETL Tools:** Apache Spark, Apache Kafka, Hadoop, Snowflake, Talend, Airflow,Data Build Tool
- **Programming Languages:** Python, SQL, Java, Scala, Shell Scripting.
- **Infrastructure as Code (IaC):** Terraform, AWS Cloud Formation, Azure Resource Manager (ARM) Templates.
- **CI/CD & DevOps:** Azure DevOps, AWS CodePipeline, Jenkins, Git, Docker, Kubernetes
- **Monitoring & Logging:** Azure Monitor, AWS CloudWatch, Google Stack driver, ELK Stack.
- **Machine Learning & AI**: Azure Machine Learning, AWS SageMaker, Google AI Platform.
- **Data Visualization & Reporting:** Power BI, Tableau, Looker, Microsoft Excel, QuickSight
- **Data Security & Governance:** Azure Security Center, AWS IAM.
- **Hadoop Components/ Big Data:** HDFS, Hive, Sqoop

EDUCATION

University of Maryland, Baltimore County, USA

- Masters / Data Science

WORK EXPERIENCE

Azure Data Engineer | Jan 2024 – Present

New York Life Insurance Company | Bethesda, Bethesda, Maryland, USA

New York Life Insurance Company (NYLIC) is the third-largest life insurance company and the largest mutual life insurance company. I am developing and maintaining scalable data models and architectures that support both current and future analytical needs. It involved in understanding business requirements and translating them into efficient data structures.

Responsibilities:

- **Developed ETL processes** using DataStage Open Studio and Apache Kafka to load and transform data from multiple sources to HDFS and Snowflake, ensuring **data quality** and governance standards.
- Implemented **RESTful and GraphQL APIs** for querying and retrieving large datasets in a structured format, ensuring data consistency.
- **Built scalable batch data pipelines** utilizing **Apache Spark (Scala)**, **Hive**, and **MapReduce**, focusing on optimization, SLA adherence, and fault tolerance for large data volumes.
- **Implemented data orchestration** using **Apache Airflow** to design, schedule, and monitor workflows, ensuring seamless data processing across systems.
- Worked on **data extraction, transformation, and aggregation** using **Spark-SQL** within **Databricks**, optimizing performance for customer behavior analysis.
- **Managed and configured resources** across the cluster using **Azure Kubernetes Service (AKS)**, enhancing scalability and resource allocation for data processing workloads.
- **Developed and maintained Snowflake** data models and schemas, including tables, views, and materialized views, to support business reporting and analytics needs.
- **Developed and maintained dbt models** to transform raw data into analytics-ready datasets, enabling faster insights for business intelligence teams.
- **Created data tables** utilizing **PyQt** for customer and policy information display and management, improving data accessibility for stakeholders.
- Experience working with **graph databases** like **Azure Cosmos DB (Gremlin API)** for graph-based data modeling and querying.
- Proficient in **container-based deployment** using **Azure Kubernetes Service (AKS)** and Docker for managing and scaling workloads.
- **Used complex SQL queries** on source systems like **Oracle 10g/11g** and **SQL Server** to perform **data profiling** and analysis, ensuring data quality and consistency.
- Designed and implemented **data integration solutions** with **Azure Data Factory**, migrating data from on-premises to cloud-based systems for enhanced scalability and efficiency.
- **Developed data warehousing solutions** with **Azure Synapse Analytics**, including building data models, ETL processes, and optimizing queries for efficient reporting and analytics.
- **Led a POC for Azure migration**, focusing on transferring on-premises data and servers to the cloud, aligning with company-wide cloud strategy.
- **Designed dbt macros and reusable models** to enhance scalability and reduce redundancy in data transformation tasks.
- Proficiency in **T-SQL** and managing **Azure SQL Database** or **Cosmos DB** for relational and NoSQL database solutions.
- Strong understanding of **REST APIs and Python scripting** for data ingestion, transformation, and integration tasks on Azure.
- **Collaborated with cross-functional teams** to implement **monitoring solutions** using **Terraform**, **Ansible**, **Docker**, and **Jenkins** to ensure reliability and maintainability of data pipelines.
- Designed and implemented **Kibana Dashboards** for real-time log analysis and monitoring, integrating **Elasticsearch** with data sources for proactive troubleshooting and transaction monitoring.
- Worked with **Jira** for tracking and managing tasks, and **Jenkins** for **CI/CD pipeline development** to streamline data pipeline deployment and updates.

Environment: Azure, Azure Synapse Analytics, Data Factory, Docker, Kubernetes, Snowflake, Apache Kafka, Spark (Scala), Hive, MapReduce, PIG, Alteryx, SQL, Tableau, Elasticsearch, Kibana, Terraform, Ansible, Jenkins, Jira, API, Cassandra, Oracle, SQL Server, Java.

AWS Data Engineer | Nov 2022 - Dec 2023
McCormick & Company | Hunt Valley, Maryland, USA

McCormick & Company is a global company that manufactures, markets, and distributes spices, seasonings, condiments, and other flavorings. I involved in designing and implementing robust Extract, Transform, Load (ETL) processes to ensure seamless integration of data from internal and external sources into the data ecosystem.

Responsibilities:

- **Developed ETL pipelines** using **PySpark** for batch processing and transformation of data across various sources.
- Utilized **AWS Lambda** and **AWS SQS** for event-driven processing and orchestration in ETL workflows.
- Worked with **Kafka** for implementing real-time data streaming solutions using **Spark Streaming** to handle large-scale data ingestion and processing.
- **Managed and optimized Spark clusters** within **Databricks**, ensuring **fault tolerance**, **high availability**, and efficient **resource utilization**.
- Used **AWS Glue** to extract, transform, and load data from **S3** to **Amazon Elasticsearch**, ensuring accurate and timely data processing.
- Implemented **Terraform** to manage infrastructure as code, automating the provisioning and configuration of **AWS resources**.
- Designed custom **APIs** to expose data processing results, enabling downstream applications and analytics platforms to access processed data.
- Developed and maintained **CI/CD pipelines** using **Jenkins**, ensuring seamless integration and deployment of data solutions.
- Migrated legacy data from on-premises data warehouses to **Snowflake**, optimizing cloud data storage and enhancing **scalability**.
- Proficiency in SQL and working with **RDS** (Postgres, MySQL) or NoSQL databases like **DynamoDB** for database management.
- Expertise in graph database modeling with **Amazon Neptune**, enabling efficient relationship-based data storage and querying.
- **Implemented version control and testing frameworks** in dbt to ensure data accuracy and streamline collaborative development.
- Skilled in stream-processing systems using **Apache Flink** on AWS and **AWS Lambda** for real-time data analytics and transformations.
- Collaborated with teams to design and implement data integration solutions, ensuring data quality, governance, and security in the cloud environment.
- **Optimized data pipelines** by integrating dbt with orchestration tools like Airflow and Azure Data Factory for seamless end-to-end workflows.
- Implemented and monitored **log management** solutions using **AWS CloudWatch** and **ELK stack (Elasticsearch, Logstash, Kibana)** for real-time monitoring and troubleshooting of data pipelines.
- Designed and developed data visualizations and dashboards using **Power BI** and **SSRS**, providing insights to business users and upper management.
- Worked on **data profiling and validation**, ensuring data quality through rigorous testing and validation processes.

Environment: AWS, Snowflake, S3, Lambda, Glue, PySpark, Spark Streaming, Kafka, Terraform, Jenkins, Power BI, SSRS, CloudWatch, Elasticsearch, Kibana, Kafka, Hive, Databricks, AWS EKS, CI/CD.

GCP Data Engineer | Jun 2021 - Jul 2022

Amgen | Hyderabad, India

Amgen is a global biopharmaceutical company dedicated to discovering, developing, and delivering innovative therapies to treat serious illnesses. My role is to continuously optimizing data pipelines for performance, reliability, and scalability. It involves in tuning database queries, optimizing data storage strategies, and implementing caching mechanisms.

Responsibilities:

- **Designed and developed ETL pipelines** for data integration, leveraging **Google Cloud Storage**, **BigQuery**, **Cloud Dataflow**, and **Dataproc** to handle large datasets and enable real-time analytics.
- **Optimized data pipelines** for performance and scalability by leveraging **PySpark** and **Spark** applications for data validation, transformation, and custom aggregations.
- Leveraged **Google Cloud Functions** and Cloud Endpoints to automate API-based data ingestion for BigQuery pipelines.
- Integrated data from various sources into **Google Cloud Storage** using **Sqoop** and processed data using **Hadoop ecosystem** tools such as **Hive** and **MapReduce**.
- Developed **reporting solutions** and **dashboards** using **SQL Server Reporting Services (SSRS)** and **Google Data Studio**, helping business users optimize their queries and understand service usage, contributing to cost-saving measures.

- Proficient in deploying and managing containers using **Google Kubernetes Engine (GKE)** and Docker for scalable microservices architecture.
- Experience with **Elasticsearch** for building data and query models, integrated with **Google Cloud Pub/Sub** for efficient data indexing and retrieval.
- Used **Apache Airflow** for managing and automating workflows in GCP, ensuring seamless execution of complex data tasks.
- Documented data lineage and dependencies using dbt to provide transparency and insights into data workflows.
- **Developed custom Python scripts** for data validation, transformations, and processing data from **Google Pub/Sub** to **BigQuery** using **Cloud Dataflow**.
- **Optimized database queries** and tuned data storage strategies, improving the overall performance of data processing tasks.
- Experience with ANSI SQL in **BigQuery** and Python-based data workflows for querying, transformation, and integration with APIs.
- Knowledge of using **Firestore** or **Spanner** for managing NoSQL and distributed relational databases in GCP.
- Coordinated with cross-functional teams to develop and deploy cloud-based data solutions, ensuring alignment with business goals and improving analytics capabilities.
- Worked on integrating **Azure Data Factory** (ADF) for seamless data movement between systems and automated **SQL activity** processing through **JSON scripts**.
- Leveraged **GCP features** such as **Google Compute Engine**, **VPC**, **IAM**, and **Cloud Load Balancing** for building scalable and secure cloud infrastructure.

Environment: GCP, BigQuery, Cloud Dataflow, Dataproc, Google Cloud Storage, Pub/Sub, Airflow, Hive, PySpark, SQL, SSRS, Python, Apache, VPC, Azure Data Factory.

Data Engineer | May 2019 - May 2021
Optum | Hyderabad, India

Optum is a health services and innovation company focused on improving healthcare delivery, offering solutions in data analytics, technology, and care management. Transformed the raw data into a structured format suitable for analysis and reporting. This often involves data cleansing, normalization, and aggregation techniques.

Responsibilities:

- Developed and optimized ETL pipelines for extracting, transforming, and loading data into cloud-based solutions like **Amazon Redshift** and **AWS S3**.
- Utilized **Apache Spark** and **PySpark** for data transformations, building scalable and efficient pipelines to process large datasets on **AWS EMR** and Hadoop clusters.
- Implemented real-time streaming applications using **Kafka**, **Apache Flink**, and **Spark Streaming** for processing high-volume, real-time data feeds.
- Automated testing and validation of data pipelines with **dbt's built-in testing capabilities**, ensuring data quality and consistency.
- Worked extensively with cloud migration, moving data from on-prem SQL Server to cloud databases such as **Amazon Redshift** and **Snowflake**.
- Designed and developed data solutions to automate data workflows, integrating **AWS Glue**, **AWS Lambda**, and **Step Functions** for seamless data pipeline orchestration.
- Leveraged **Python** and **Scala** for building data processing scripts, improving performance and data integrity across large datasets.
- Built and maintained CI/CD pipelines with **Jenkins** and **Docker**, automating the deployment of custom applications and managing containerized environments with **Kubernetes**.
- Optimized SQL queries and leveraged **Hive** and **Spark SQL** to enhance the performance of data processing algorithms and ensure data accuracy.
- Designed and documented business requirements for data-related projects, ensuring alignment with SDLC methodology for consistent and efficient project execution.

Environment: AWS, Amazon Redshift, AWS S3, AWS Glue, AWS Lambda, Step Functions, ETL, Kafka, PySpark, Scala, Snowflake, SQL, Spark, Docker, Kubernetes, Jenkins, Hive, Apache Flink, Python, CI/CD, Git, Linux